Definition 1. Let Var be the set of propositional variables. Then ϕ is defined inductively as follows:

$$\phi ::= p \mid \phi \vee \phi \mid X\phi \mid \phi U \psi \mid \langle a \rangle \phi$$

Definition 2. Let ϕ be a formula and $p \in Var$. Then the Closure of ϕ is inductively defined as

$$\begin{aligned} Cl(p) &= Cl(\neg p) = \{p, \neg p\} \\ Cl(\neg\phi) &= Cl(\phi) \\ Cl(\phi \vee \psi) &= \{\phi \vee \psi, \neg(\phi \vee \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(X\phi) &= \{X\phi, \neg X\phi\} \cup Cl(\phi) \\ Cl(\phi U \psi) &= \{\phi U \psi, \neg(\phi U \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\ Cl(\langle a \rangle \phi) &= \{\langle a \rangle \phi, \neg \langle a \rangle \phi\} \cup Cl(\phi) \end{aligned}$$

Definition 3. An atom A is a subset of $Cl(\phi)$ such that

$$\begin{array}{ll} \text{if } \psi \in Cl(\phi), & \text{then } \psi \in A \text{ iff } \neg\psi \notin A, \\ \text{if } \phi_1 \vee \phi_2 \in Cl(\phi), & \text{then } \phi_1 \vee \phi_2 \in A \text{ iff } \phi_1 \in A \text{ or } \phi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \text{then if } \phi_1 U \phi_2 \in A \text{ then } \psi_1 \in A \text{ or } \psi_2 \in A \\ \text{if } \phi_1 U \phi_2 \in Cl(\phi), & \text{then if } \phi_2 \in A \text{ then } \phi_1 U \phi_2 \in A \end{array}$$

Definition 4. We say two atoms A, B are in relation $R_x (AR_xB)$ iff

$$\begin{array}{ll} \text{if } X\psi \in Cl(\phi), & \text{then } X\psi \in A \text{ iff } \psi \in B, \\ \text{if } \psi_1 U \psi_2 \in Cl(\phi), & \text{then if } \psi_1 U \psi_2, \neg\psi_2 \in A, \text{ then } \psi_1 U \psi_2 \in B, \\ & \text{and if } \psi_1 U \psi_2 \in B, \psi_1 \in A, \text{ then } \psi_1 U \psi_2 \in A. \end{array}$$

Lemma 5. *For any atom A there exists an atom B s.t. AR_xB.*

Proof. Assume A is an atom. Then we define B' as follows:

$$B' = \{\phi \mid X\phi \in A\} \cup \{\neg\phi \mid \neg X\phi \in A\} \cup \{\psi_1 U \psi_2 \mid \psi_1 U \psi_2, \neg\psi_2 \in A\} \cup \{\neg(\psi_1 U \psi_2) \mid \neg(\psi_1 U \psi_2), \psi_1 \in A\}$$

Now we show that B is consistent. Assume it is not. Then there exists $\alpha_1, ..., \alpha_n$ s.t. $B \vdash \alpha_1 \wedge \alpha_2 ... \wedge \alpha_n \to \bot$. Let's define $\hat{B} = \bigwedge B$. It follows:

$$\begin{array}{ll} \vdash \hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n \to \bot & \\ \vdash X(\hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n \to \bot) & \text{By X-necessitation} \\ \vdash X(\hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n) \to X\bot & \text{F2} \\ \vdash X(\hat{B}) \wedge X\alpha_1 \wedge ... \wedge X\alpha_n \to X\bot & \end{array}$$

Assume, $a_i = \phi$. Then by definition we have $A \vdash Xa_i$. If $a_i = \neg\phi$, then we can use F1 to obtain $A \vdash X\neg\phi$. If $a_i = \psi_1 U \psi_2$ then by definition, $\psi_1 U \psi_2 \in A$ and $\neg\psi_2 \in A$. Consider the following proof steps:

$A \vdash \neg\psi_2$	
$A \vdash \psi_1 U \psi_2$	
$A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2$	
$A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \neg\psi_2$	Unfolding U
$A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_1 \wedge X(\psi_1 U \psi_2))$	distributivity and other rules
$A \vdash \psi_1 \wedge X(\psi_1 U \psi_2)$	
$A \vdash X(\psi_1 U \psi_2)$.	

Hence, $A \vdash Xa_i$. If $a_i = \neg(\psi_1 U \psi_2)$, then by definition, $\neg(\psi_1 U \psi_2), \psi_1 \in A$. Our goal is to derive $A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1 \rightarrow X\neg(\psi_1 U \psi_2)$.

$A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1$	
$A \vdash \neg(\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \psi_1$	Unfolding U
$A \vdash \neg\psi_2 \wedge (\neg\psi_1 \vee \neg X(\psi_1 U \psi_2)) \wedge \psi_1$	
$A \vdash \neg\psi_1 \vee \neg X(\psi_1 U \psi_2)$	
$\neg X(\psi_1 U \psi_2)$	ψ_1 is derivable
$A \vdash X\neg(\psi_1 U \psi_2)$	F1

By similar argument, we can conclude $A \vdash X\hat{B}$. So it follows

$$A \vdash X\bot$$

Because $X\bot \rightarrow \bot$ is a tautology, we conclude $A \vdash \bot$ which is a contradiction. Now, by the Extension Lemma, we can extend B' to an atom B. It is easy to check, that AR_xB.

□

Lemma 6. *Let $\Gamma = \{\phi\}'$ be a consistent set of formulas. Then Γ' can be extended to an atom.*

Proof. Let $n = |CL(\phi)|$ and consider the enumeration $\psi_1, \psi_2, ..., \psi_n$ of $CL(\phi)$. We define $\Gamma_0 = \{\phi\}$ and then inductively

$$\Gamma_{n+1} := \begin{cases} \Gamma_n \cup \psi_{n+1}, & \text{if } \Gamma_n \cup \psi_{n+1} \text{ is consistent,} \\ \Gamma_n \cup \neg\psi_{n+1}, & \text{else .} \end{cases}$$

Then $\Gamma^+ = \bigcup_n \Gamma_n$ is our atom. Idea: Show for each Γ_n is consistent for each n.

□

Lemma 7. *Assume* $\psi_1 U \psi_2 \in A_i$. *Then there is an* $n \in \mathbb{N}$ *with* $A_i R_x A_{i+1} R_x ... R_x A_n$ *s.t.* $\psi_2 \in A_n$ *and for all* $i \leq j < n : \psi_1 \in A_j$.